In [188…
```
import pandas as pd
```

In [189…
```
d = pd.read_csv(r"C:\Users\ADMIN\Documents\Duration  Date  Pulse  Maxp.txt")
```

In [190…
```
d.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 32 entries, 0 to 31
Data columns (total 5 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0    Duration  32 non-null     int64
 1   Date        31 non-null     object
 2   Pulse       32 non-null     int64
 3   Maxpulse    32 non-null     int64
 4   Calories    30 non-null     float64
dtypes: float64(1), int64(3), object(1)
memory usage: 1.4+ KB
```

In [191…
```
d.columns
```

Out[191…
```
Index([' Duration', 'Date', 'Pulse', 'Maxpulse', 'Calories'], dtype='object')
```

In [ ]:

In [192…
```
d.isna().sum()
```

Out[192…
```
 Duration    0
Date         1
Pulse        0
Maxpulse     0
Calories     2
dtype: int64
```

In [193…
```
d.isna().any()
```

Out[193…
```
 Duration    False
Date          True
Pulse        False
Maxpulse     False
Calories      True
dtype: bool
```

In [194…
```
# to view rows with missing values
d.isna().any(axis = 1)
```

Out[194…

```
0     False
1     False
2     False
3     False
4     False
5     False
6     False
7     False
8     False
9     False
10    False
11    False
12    False
13    False
14    False
15    False
16    False
17    False
18     True
19    False
20    False
21    False
22     True
23    False
24    False
25    False
26    False
27    False
28     True
29    False
30    False
31    False
dtype: bool
```

In [195…

```
d.isna().any()
```

Out[195…

```
 Duration    False
Date          True
Pulse        False
Maxpulse     False
Calories      True
dtype: bool
```

In [196…

```
d[d.isna().any(axis = 1)]
```

Out[196…

	Duration	Date	Pulse	Maxpulse	Calories
18	45	'2020/12/18'	90	112	NaN
22	45	NaN	100	119	282.0
28	60	'2020/12/28'	103	132	NaN

In [197…

```
d.dropna()
```

Out[197…

	Duration	Date	Pulse	Maxpulse	Calories
0	60	'2020/12/01'	110	130	409.1
1	60	'2020/12/02'	117	145	479.0
2	60	'2020/12/03'	103	135	340.0
3	45	'2020/12/04'	109	175	282.4
4	45	'2020/12/05'	117	148	406.0
5	60	'2020/12/06'	102	127	300.0
6	60	'2020/12/07'	110	136	374.0
7	450	'2020/12/08'	104	134	253.3
8	30	'2020/12/09'	109	133	195.1
9	60	'2020/12/10'	98	124	269.0
10	60	'2020/12/11'	103	147	329.3
11	60	'2020/12/12'	100	120	250.7
12	60	'2020/12/12'	100	120	250.7
13	60	'2020/12/13'	106	128	345.3
14	60	'2020/12/14'	104	132	379.3
15	60	'2020/12/15'	98	123	275.0
16	60	'2020/12/16'	98	120	215.2
17	60	'2020/12/17'	100	120	300.0
19	60	'2020/12/19'	103	123	323.0
20	45	'2020/12/20'	97	125	243.0
21	60	'2020/12/21'	108	131	364.2
23	60	'2020/12/23'	130	101	300.0
24	45	'2020/12/24'	105	132	246.0
25	60	'2020/12/25'	102	126	334.5
26	60	2020/12/26	100	120	250.0
27	60	'2020/12/27'	92	118	241.0
29	60	'2020/12/29'	100	132	280.0
30	60	'2020/12/30'	102	129	380.3
31	60	'2020/12/31'	92	115	243.0

In [198…

```
d.fillna(23) # all the missing values in your data
```

Out[198...

	Duration	Date	Pulse	Maxpulse	Calories
0	60	'2020/12/01'	110	130	409.1
1	60	'2020/12/02'	117	145	479.0
2	60	'2020/12/03'	103	135	340.0
3	45	'2020/12/04'	109	175	282.4
4	45	'2020/12/05'	117	148	406.0
5	60	'2020/12/06'	102	127	300.0
6	60	'2020/12/07'	110	136	374.0
7	450	'2020/12/08'	104	134	253.3
8	30	'2020/12/09'	109	133	195.1
9	60	'2020/12/10'	98	124	269.0
10	60	'2020/12/11'	103	147	329.3
11	60	'2020/12/12'	100	120	250.7
12	60	'2020/12/12'	100	120	250.7
13	60	'2020/12/13'	106	128	345.3
14	60	'2020/12/14'	104	132	379.3
15	60	'2020/12/15'	98	123	275.0
16	60	'2020/12/16'	98	120	215.2
17	60	'2020/12/17'	100	120	300.0
18	45	'2020/12/18'	90	112	23.0
19	60	'2020/12/19'	103	123	323.0
20	45	'2020/12/20'	97	125	243.0
21	60	'2020/12/21'	108	131	364.2
22	45	23	100	119	282.0
23	60	'2020/12/23'	130	101	300.0
24	45	'2020/12/24'	105	132	246.0
25	60	'2020/12/25'	102	126	334.5
26	60	2020/12/26	100	120	250.0
27	60	'2020/12/27'	92	118	241.0
28	60	'2020/12/28'	103	132	23.0
29	60	'2020/12/29'	100	132	280.0

	Duration	Date	Pulse	Maxpulse	Calories
30	60	'2020/12/30'	102	129	380.3
31	60	'2020/12/31'	92	115	243.0

In [199…

```
d.head()
```

Out[199…

	Duration	Date	Pulse	Maxpulse	Calories
0	60	'2020/12/01'	110	130	409.1
1	60	'2020/12/02'	117	145	479.0
2	60	'2020/12/03'	103	135	340.0
3	45	'2020/12/04'	109	175	282.4
4	45	'2020/12/05'	117	148	406.0

In [200…

```
d["Date"] = d["Date"].fillna("'2020/12/22'")
```

In [201…

```
d["Calories"].fillna(300.0)
```

Out[201…

```
0     409.1
1     479.0
2     340.0
3     282.4
4     406.0
5     300.0
6     374.0
7     253.3
8     195.1
9     269.0
10    329.3
11    250.7
12    250.7
13    345.3
14    379.3
15    275.0
16    215.2
17    300.0
18    300.0
19    323.0
20    243.0
21    364.2
22    282.0
23    300.0
24    246.0
25    334.5
26    250.0
27    241.0
28    300.0
29    280.0
30    380.3
31    243.0
Name: Calories, dtype: float64
```

In [202…

```
d["Calories"].mode()
```

Out[202…

```
0    300.0
Name: Calories, dtype: float64
```

In [203…

```
d.head()
```

Out[203…

	Duration	Date	Pulse	Maxpulse	Calories
0	60	'2020/12/01'	110	130	409.1
1	60	'2020/12/02'	117	145	479.0
2	60	'2020/12/03'	103	135	340.0
3	45	'2020/12/04'	109	175	282.4
4	45	'2020/12/05'	117	148	406.0

In [204…

```
# loc and iloc - label location and integer location
```

In [205…
```
d.iloc[2,2]
```
Out[205…
```
103
```

In [206…
```
d.loc[2,"Pulse"]
```
Out[206…
```
103
```

In [207…
```
d.iloc[18,4] = 304
```

In [208…
```
d.iloc[18,4]
```
Out[208…
```
304.0
```

In [209…
```
d.loc[28,"Calories"]
```
Out[209…
```
nan
```

In [210…
```
d.loc[28,"Calories"] = 300
```

In [211…
```
d.loc[28,"Calories"]
```
Out[211…
```
300.0
```

In [212…
```
d.Calories.isna().any()
```
Out[212…
```
False
```

In [213…
```
d.iloc[26,1] = "'2020/12/26'"
```

In [214…
```
d["Date"] = pd.to_datetime(d["Date"])
```

In [215…
```
d.info()
```
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 32 entries, 0 to 31
Data columns (total 5 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0    Duration  32 non-null     int64
 1   Date       32 non-null     datetime64[ns]
 2   Pulse      32 non-null     int64
 3   Maxpulse   32 non-null     int64
 4   Calories   32 non-null     float64
dtypes: datetime64[ns](1), float64(1), int64(3)
memory usage: 1.4 KB
```

In [216…
```
d.duplicated().any()
```
Out[216…
```
True
```

In [217…
```
d.duplicated().sum()
```

Out[217… 1

In [218…
```
d[d.duplicated()]
```

Out[218…

	Duration	Date	Pulse	Maxpulse	Calories
12	60	2020-12-12	100	120	250.7

In [219…
```
d[d.duplicated(keep = False)]
```

Out[219…

	Duration	Date	Pulse	Maxpulse	Calories
11	60	2020-12-12	100	120	250.7
12	60	2020-12-12	100	120	250.7

In [220…
```
d = d.drop_duplicates()
```

In [221…
```
d.reset_index()
```

Out[221…

	index	Duration	Date	Pulse	Maxpulse	Calories
0	0	60	2020-12-01	110	130	409.1
1	1	60	2020-12-02	117	145	479.0
2	2	60	2020-12-03	103	135	340.0
3	3	45	2020-12-04	109	175	282.4
4	4	45	2020-12-05	117	148	406.0
5	5	60	2020-12-06	102	127	300.0
6	6	60	2020-12-07	110	136	374.0
7	7	450	2020-12-08	104	134	253.3
8	8	30	2020-12-09	109	133	195.1
9	9	60	2020-12-10	98	124	269.0
10	10	60	2020-12-11	103	147	329.3
11	11	60	2020-12-12	100	120	250.7
12	13	60	2020-12-13	106	128	345.3
13	14	60	2020-12-14	104	132	379.3
14	15	60	2020-12-15	98	123	275.0
15	16	60	2020-12-16	98	120	215.2
16	17	60	2020-12-17	100	120	300.0
17	18	45	2020-12-18	90	112	304.0
18	19	60	2020-12-19	103	123	323.0
19	20	45	2020-12-20	97	125	243.0
20	21	60	2020-12-21	108	131	364.2
21	22	45	2020-12-22	100	119	282.0
22	23	60	2020-12-23	130	101	300.0
23	24	45	2020-12-24	105	132	246.0
24	25	60	2020-12-25	102	126	334.5
25	26	60	2020-12-26	100	120	250.0
26	27	60	2020-12-27	92	118	241.0
27	28	60	2020-12-28	103	132	300.0
28	29	60	2020-12-29	100	132	280.0
29	30	60	2020-12-30	102	129	380.3

	index	Duration	Date	Pulse	Maxpulse	Calories
30	31	60	2020-12-31	92	115	243.0

In [222…
```
d.reset_index(drop = True, inplace = True)
```

In [223…
```
d[" Duration"].value_counts()
```

Out[223…
```
 Duration
60     23
45      6
450     1
30      1
Name: count, dtype: int64
```

In [224…
```
d.iloc[7, 0]
```

Out[224…
```
450
```

In [225…
```
d.iloc[7, 0] = 45
```

In [226…
```
d.groupby(" Duration")["Calories"].mean().round(1)
```

Out[226…
```
 Duration
30    195.1
45    288.1
60    316.6
Name: Calories, dtype: float64
```

In [227…
```
d.groupby(" Duration")["Calories"].mean().reset_index()
```

Out[227…

	Duration	Calories
0	30	195.100000
1	45	288.100000
2	60	316.604348

In [228…
```
d[" Duration"].corr(d["Calories"])
```

Out[228…
```
0.34342412002455314
```

In [229…
```
# + - 0 to 1
```

In [230…
```
d.corr()
```

Out[230…

	Duration	Date	Pulse	Maxpulse	Calories
Duration	1.000000	0.173344	-0.059452	-0.250033	0.343424
Date	0.173344	1.000000	-0.375585	-0.525927	-0.366317
Pulse	-0.059452	-0.375585	1.000000	0.269672	0.482506
Maxpulse	-0.250033	-0.525927	0.269672	1.000000	0.335191
Calories	0.343424	-0.366317	0.482506	0.335191	1.000000

In [231…

```
a = pd.read_excel(r"C:\Users\ADMIN\Desktop\Sage's PowerBI BootCamp\automobile_data
```

In [232…

```
a.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 203 entries, 0 to 202
Data columns (total 20 columns):
 #   Column             Non-Null Count  Dtype  
---  ------             --------------  -----  
 0   make               203 non-null    object 
 1   fuel_type          203 non-null    object 
 2   num_of_doors       201 non-null    object 
 3   body_style         203 non-null    object 
 4   drive_wheels       203 non-null    object 
 5   engine_location    203 non-null    object 
 6   wheel_base         203 non-null    float64
 7   length             203 non-null    float64
 8   width              203 non-null    float64
 9   height             203 non-null    float64
 10  curb_weight        203 non-null    int64  
 11  engine_type        203 non-null    object 
 12  num_of_cylinders   203 non-null    object 
 13  engine_size        203 non-null    int64  
 14  fuel_system        203 non-null    object 
 15  compression_ratio  203 non-null    float64
 16  horsepower         203 non-null    int64  
 17  city_mpg           203 non-null    int64  
 18  highway_mpg        203 non-null    int64  
 19  price              203 non-null    int64  
dtypes: float64(5), int64(6), object(9)
memory usage: 31.8+ KB
```

In [233…

```
a.duplicated().any()
```

Out[233…

```
True
```

In [234…

```
a[a.duplicated(keep = False)]
```

Out[234…

	make	fuel_type	num_of_doors	body_style	drive_wheels	engine_location	wheel_b
87	mitsubishi	gas	four	sedan	fwd	front	
88	mitsubishi	gas	four	sedan	fwd	front	

In [235…

```
a.make.nunique()
```

Out[235…

```
21
```

In [236…

```
a.groupby("fuel_type")["make"].value_counts()
```

Out[236…

```
fuel_type  make
diesel     peugot            5
           volkswagen        4
           mercedes-benz     4
           toyota            3
           mazda             2
           volvo             1
           nissan            1
gas        toyota           29
           nissan           17
           mazda            15
           mitsubishi       13
           honda            13
           subaru           12
           volvo            10
           dodge             9
           bmw               8
           volkswagen        8
           audi              7
           plymouth          7
           saab              6
           peugot            6
           porsche           5
           isuzu             4
           mercedes-benz     4
           jaguar            3
           chevrolet         3
           alfa-romero       3
           mercury           1
Name: count, dtype: int64
```

In [237…

```
pd.crosstab(a["fuel_type"], a["make"])
```

Out[237…

make	alfa-romero	audi	bmw	chevrolet	dodge	honda	isuzu	jaguar	mazda	mercedes ben
fuel_type										
diesel	0	0	0	0	0	0	0	0	2	
gas	3	7	8	3	9	13	4	3	15	

2 rows × 21 columns

In [238…

```
a.body_style.value_counts()
```

Out[238…

```
body_style
sedan          96
hatchback      69
wagon          24
hardtop         8
convertible     6
Name: count, dtype: int64
```

In [239…

```
a.head()
```

Out[239…

	make	fuel_type	num_of_doors	body_style	drive_wheels	engine_location	wheel_base
0	alfa-romero	gas	two	convertible	rwd	front	88.6
1	alfa-romero	gas	two	convertible	rwd	front	88.6
2	alfa-romero	gas	two	hatchback	rwd	front	94.5
3	audi	gas	four	sedan	fwd	front	99.8
4	audi	gas	four	sedan	4wd	front	99.4

In [240…

```
a.groupby("engine_type")["engine_size"].mean()
```

Out[240…

```
engine_type
dohc     157.333333
dohcv    203.000000
l        129.583333
ohc      117.342466
ohcf     124.466667
ohcv     216.692308
rotor     72.500000
Name: engine_size, dtype: float64
```

In [241…

```
a.engine_type.value_counts()
```

Out[241…

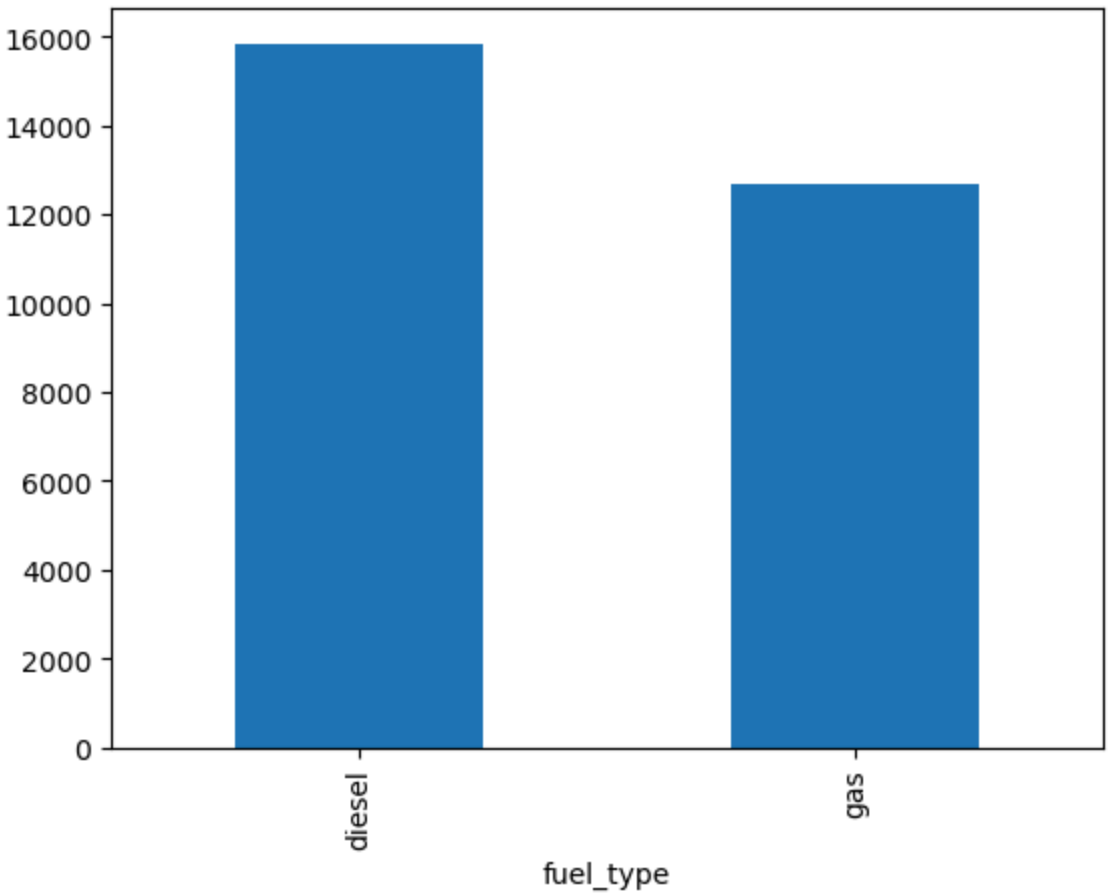
```
engine_type
ohc       146
ohcf       15
ohcv       13
dohc       12
l          12
rotor       4
dohcv       1
Name: count, dtype: int64
```

In [242…

```
a.groupby("fuel_type")["price"].mean().plot.bar()
```

Out[242…

```
<Axes: xlabel='fuel_type'>
```



In [243…

```
a.groupby("make")["price"].agg(["sum", "mean", "count"]).sort_values(ascending = Fa
```

Out[243…

make	sum	mean	count
jaguar	103800	34600.000000	3
mercedes-benz	269176	33647.000000	8
bmw	208950	26118.750000	8
porsche	125602	25120.400000	5
volvo	198695	18063.181818	11
mercury	16503	16503.000000	1
alfa-romero	46495	15498.333333	3
peugot	170380	15489.090909	11
audi	107155	15307.857143	7
saab	91340	15223.333333	6
mazda	181099	10652.882353	17
nissan	187482	10415.666667	18
volkswagen	120930	10077.500000	12
toyota	316346	9885.812500	32
mitsubishi	120117	9239.769231	13
subaru	102495	8541.250000	12
honda	106401	8184.692308	13
plymouth	55744	7963.428571	7
dodge	70879	7875.444444	9
chevrolet	18021	6007.000000	3
isuzu	17833	4458.250000	4

In [244…

```
a.groupby(["make", "horsepower"])["engine_location"].value_counts()
```

Out[244…

make	horsepower	engine_location	
alfa-romero	111	front	2
	154	front	1
audi	102	front	1
	110	front	3
	115	front	1
	140	front	1
	160	front	1
bmw	101	front	2
	121	front	3
	182	front	3
chevrolet	48	front	1
	70	front	2
dodge	68	front	5
	88	front	1
	102	front	2
	145	front	1
honda	58	front	1
	60	front	1
	76	front	5
	86	front	4
	100	front	1
	101	front	1
isuzu	70	front	2
	78	front	1
	90	front	1
jaguar	176	front	2
	262	front	1
mazda	64	front	1
	68	front	5
	72	front	1
	84	front	5
	101	front	3
	120	front	1
	135	front	1
mercedes-benz	123	front	4
	155	front	2
	184	front	2
mercury	175	front	1
mitsubishi	68	front	3
	88	front	3
	102	front	1
	116	front	3
	145	front	3
nissan	55	front	1
	69	front	9
	97	front	2
	152	front	3
	160	front	2
	200	front	1
peugot	95	front	7
	97	front	3
	142	front	1
plymouth	68	front	4
	88	front	1
	102	front	1

```
                145         front              1
porsche         143         front              1
                207         rear               3
                288         front              1
saab            110         front              4
                160         front              2
subaru          69          front              1
                73          front              2
                82          front              5
                94          front              2
                111         front              2
toyota          56          front              2
                62          front              6
                70          front              7
                73          front              1
                92          front              4
                112         front              2
                116         front              6
                156         front              2
                161         front              2
volkswagen      52          front              2
                68          front              2
                85          front              3
                88          front              1
                90          front              2
                100         front              1
                110         front              1
volvo           106         front              1
                114         front              6
                134         front              1
                160         front              1
                162         front              2
Name: count, dtype: int64
```

In [245…

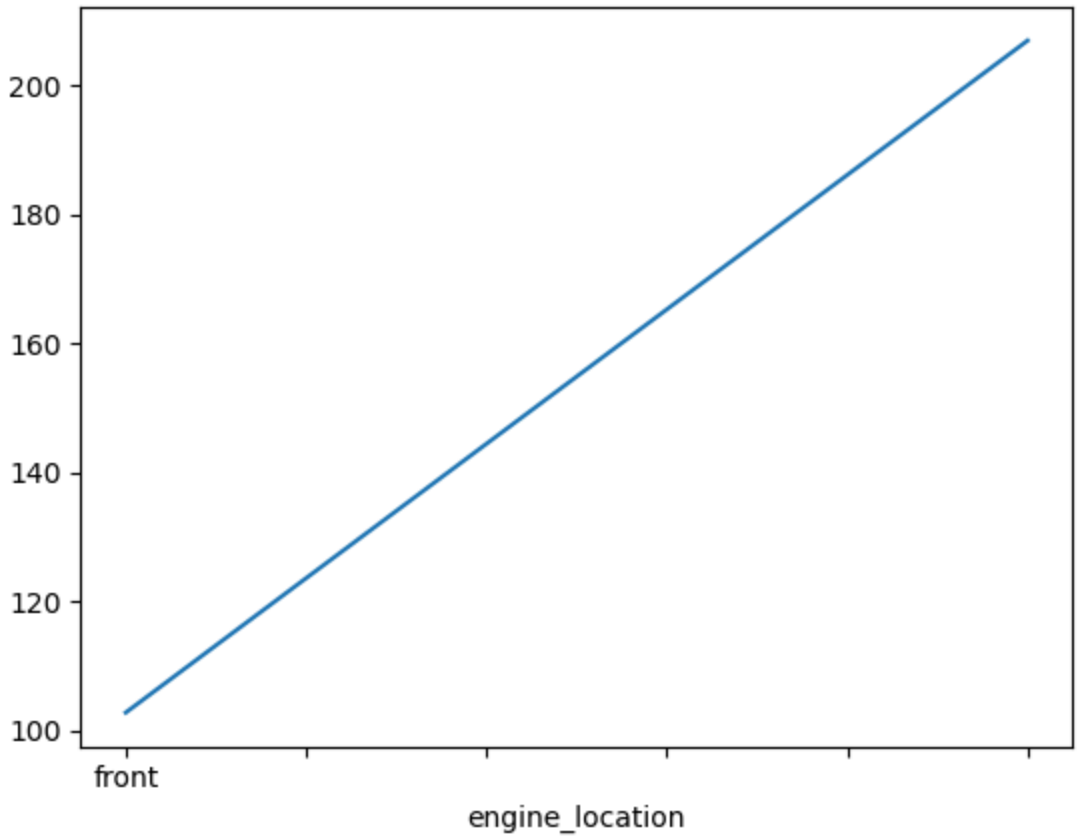
```
pd.set_option("display.max_rows", 96)
```

In [246…

```
a.groupby(["engine_location"])["horsepower"].mean().plot()
```

Out[246…

```
<Axes: xlabel='engine_location'>
```



In [247…

```
a.drive_wheels.value_counts()
```

Out[247…

```
drive_wheels
fwd     118
rwd      76
4wd       5
4wd       4
Name: count, dtype: int64
```

In [248…

```
a[a.drive_wheels == "4wd "]
```

Out[248…

	make	fuel_type	num_of_doors	body_style	drive_wheels	engine_location	wheel_ba
4	audi	gas	four	sedan	4wd	front	99
9	audi	gas	two	hatchback	4wd	front	99
138	subaru	gas	two	hatchback	4wd	front	93
142	subaru	gas	four	sedan	4wd	front	97
143	subaru	gas	four	sedan	4wd	front	97

In [249…

```
a[a.drive_wheels == "4wd"]
```

Out[249…

	make	fuel_type	num_of_doors	body_style	drive_wheels	engine_location	wheel_ba
146	subaru	gas	four	wagon	4wd	front	96
147	subaru	gas	four	wagon	4wd	front	96
152	toyota	gas	four	wagon	4wd	front	95
153	toyota	gas	four	wagon	4wd	front	95

In [250…

```
a.groupby("drive_wheels")[["city_mpg","highway_mpg"]].mean().round(1)
```

Out[250…

drive_wheels	city_mpg	highway_mpg
4wd	25.0	29.0
4wd	21.6	25.8
fwd	28.4	34.3
rwd	20.6	25.7

In [251…

```
a.drive_wheels.str.strip()
```

Out[251…

```
0      rwd
1      rwd
2      rwd
3      fwd
4      4wd
      ...
198    rwd
199    rwd
200    rwd
201    rwd
202    rwd
Name: drive_wheels, Length: 203, dtype: object
```

In [252…

```
a.drive_wheels = a.drive_wheels.str.strip()
```

In [253…

```
a.drive_wheels.value_counts()
```

Out[253…

```
drive_wheels
fwd    118
rwd     76
4wd      9
Name: count, dtype: int64
```

In [254…

```
a[a.isna().any(axis = 1)]
```

Out[254…

	make	fuel_type	num_of_doors	body_style	drive_wheels	engine_location	wheel_base
27	dodge	gas	NaN	sedan	fwd	front	93.7
63	mazda	diesel	NaN	sedan	fwd	front	98.8

In [255…

```
a[a.body_style == "sedan"]
```

Out[255...

	make	fuel_type	num_of_doors	body_style	drive_wheels	engine_location	whee
3	audi	gas	four	sedan	fwd	front	
4	audi	gas	four	sedan	4wd	front	
5	audi	gas	two	sedan	fwd	front	
6	audi	gas	four	sedan	fwd	front	
8	audi	gas	four	sedan	fwd	front	
10	bmw	gas	two	sedan	rwd	front	
11	bmw	gas	four	sedan	rwd	front	
12	bmw	gas	two	sedan	rwd	front	
13	bmw	gas	four	sedan	rwd	front	
14	bmw	gas	four	sedan	rwd	front	
15	bmw	gas	four	sedan	rwd	front	
16	bmw	gas	two	sedan	rwd	front	
17	bmw	gas	four	sedan	rwd	front	
20	chevrolet	gas	four	sedan	fwd	front	
25	dodge	gas	four	sedan	fwd	front	
26	dodge	gas	four	sedan	fwd	front	
27	dodge	gas	NaN	sedan	fwd	front	
35	honda	gas	four	sedan	fwd	front	
39	honda	gas	four	sedan	fwd	front	
40	honda	gas	four	sedan	fwd	front	
41	honda	gas	four	sedan	fwd	front	
42	honda	gas	two	sedan	fwd	front	
43	isuzu	gas	four	sedan	rwd	front	
44	isuzu	gas	two	sedan	fwd	front	
45	isuzu	gas	four	sedan	fwd	front	
47	jaguar	gas	four	sedan	rwd	front	
48	jaguar	gas	four	sedan	rwd	front	
49	jaguar	gas	two	sedan	rwd	front	
53	mazda	gas	four	sedan	fwd	front	
54	mazda	gas	four	sedan	fwd	front	

	make	fuel_type	num_of_doors	body_style	drive_wheels	engine_location	whee
60	mazda	gas	four	sedan	fwd	front	
62	mazda	gas	four	sedan	fwd	front	
63	mazda	diesel	NaN	sedan	fwd	front	
65	mazda	gas	four	sedan	rwd	front	
66	mazda	diesel	four	sedan	rwd	front	
67	mercedes-benz	diesel	four	sedan	rwd	front	
70	mercedes-benz	diesel	four	sedan	rwd	front	
71	mercedes-benz	gas	four	sedan	rwd	front	
73	mercedes-benz	gas	four	sedan	rwd	front	
85	mitsubishi	gas	four	sedan	fwd	front	
86	mitsubishi	gas	four	sedan	fwd	front	
87	mitsubishi	gas	four	sedan	fwd	front	
88	mitsubishi	gas	four	sedan	fwd	front	
89	nissan	gas	two	sedan	fwd	front	
90	nissan	diesel	two	sedan	fwd	front	
91	nissan	gas	two	sedan	fwd	front	
92	nissan	gas	four	sedan	fwd	front	
94	nissan	gas	two	sedan	fwd	front	
96	nissan	gas	four	sedan	fwd	front	
100	nissan	gas	four	sedan	fwd	front	
101	nissan	gas	four	sedan	fwd	front	
103	nissan	gas	four	sedan	fwd	front	
107	peugot	gas	four	sedan	rwd	front	
108	peugot	diesel	four	sedan	rwd	front	
111	peugot	gas	four	sedan	rwd	front	
112	peugot	diesel	four	sedan	rwd	front	
115	peugot	gas	four	sedan	rwd	front	
116	peugot	diesel	four	sedan	rwd	front	

	make	fuel_type	num_of_doors	body_style	drive_wheels	engine_location	whee
117	peugot	gas	four	sedan	rwd	front	
121	plymouth	gas	four	sedan	fwd	front	
122	plymouth	gas	four	sedan	fwd	front	
131	saab	gas	four	sedan	fwd	front	
133	saab	gas	four	sedan	fwd	front	
135	saab	gas	four	sedan	fwd	front	
139	subaru	gas	four	sedan	fwd	front	
140	subaru	gas	four	sedan	fwd	front	
141	subaru	gas	four	sedan	fwd	front	
142	subaru	gas	four	sedan	4wd	front	
143	subaru	gas	four	sedan	4wd	front	
154	toyota	gas	four	sedan	fwd	front	
156	toyota	diesel	four	sedan	fwd	front	
158	toyota	gas	four	sedan	fwd	front	
160	toyota	gas	four	sedan	fwd	front	
161	toyota	gas	two	sedan	rwd	front	
163	toyota	gas	two	sedan	rwd	front	
171	toyota	gas	four	sedan	fwd	front	
172	toyota	diesel	four	sedan	fwd	front	
174	toyota	gas	four	sedan	fwd	front	
178	toyota	gas	four	sedan	rwd	front	
180	volkswagen	diesel	two	sedan	fwd	front	
181	volkswagen	gas	two	sedan	fwd	front	
182	volkswagen	diesel	four	sedan	fwd	front	
183	volkswagen	gas	four	sedan	fwd	front	
184	volkswagen	gas	four	sedan	fwd	front	
185	volkswagen	diesel	four	sedan	fwd	front	
186	volkswagen	gas	four	sedan	fwd	front	
189	volkswagen	gas	four	sedan	fwd	front	
190	volkswagen	diesel	four	sedan	fwd	front	

	make	fuel_type	num_of_doors	body_style	drive_wheels	engine_location	whee
192	volvo	gas	four	sedan	rwd	front	
194	volvo	gas	four	sedan	rwd	front	
196	volvo	gas	four	sedan	rwd	front	
198	volvo	gas	four	sedan	rwd	front	
199	volvo	gas	four	sedan	rwd	front	
200	volvo	gas	four	sedan	rwd	front	
201	volvo	diesel	four	sedan	rwd	front	
202	volvo	gas	four	sedan	rwd	front	

In [256…

```
a[a.duplicated(keep = False)]
```

Out[256…

	make	fuel_type	num_of_doors	body_style	drive_wheels	engine_location	wheel_b
87	mitsubishi	gas	four	sedan	fwd	front	
88	mitsubishi	gas	four	sedan	fwd	front	

In [257…

```
a = a.drop_duplicates().reset_index(drop = True)
```

In [258…

```
a
```

Out[258…

	make	fuel_type	num_of_doors	body_style	drive_wheels	engine_location	wheel_ba
0	alfa-romero	gas	two	convertible	rwd	front	88
1	alfa-romero	gas	two	convertible	rwd	front	88
2	alfa-romero	gas	two	hatchback	rwd	front	94
3	audi	gas	four	sedan	fwd	front	99
4	audi	gas	four	sedan	4wd	front	99
...	...	...	...	...	...	...	
197	volvo	gas	four	sedan	rwd	front	109
198	volvo	gas	four	sedan	rwd	front	109
199	volvo	gas	four	sedan	rwd	front	109
200	volvo	diesel	four	sedan	rwd	front	109
201	volvo	gas	four	sedan	rwd	front	109

202 rows × 20 columns

In [ ]: